

AI ASSISTED CODING ASSIGNMENT

– 3.5

CH.Pooja

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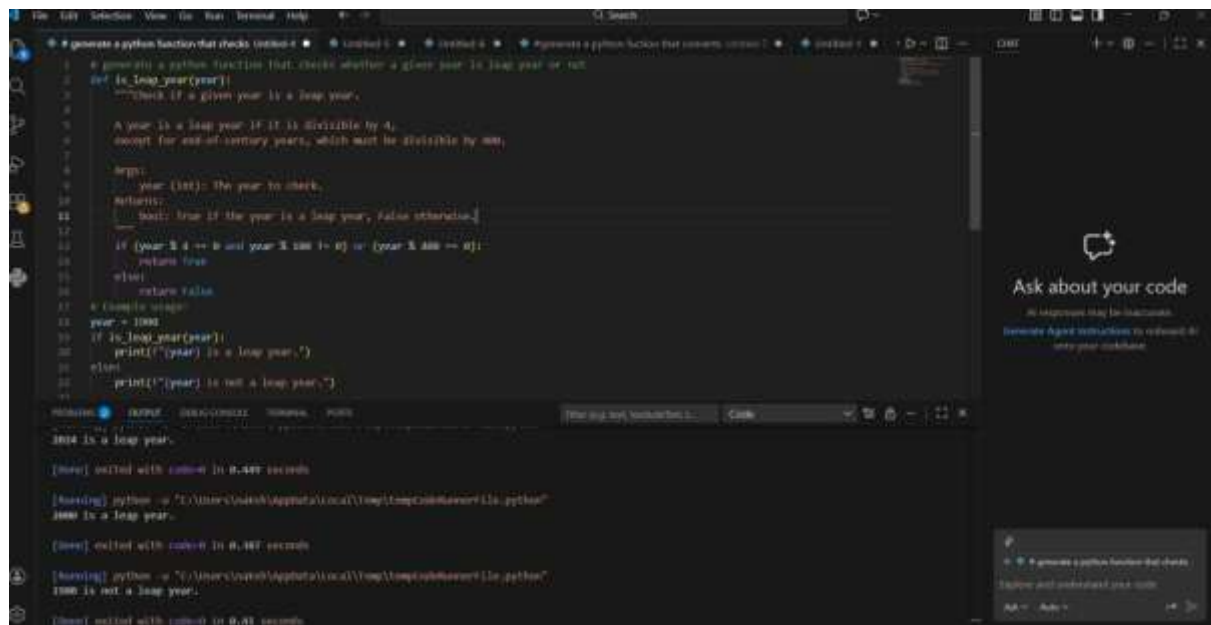
BATCH-03

Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Week2 - Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.



```
1 # generate a python function that checks whether a given year is leap year or not
2 def is_leap_year(year):
3     """Check if a given year is a leap year.
4
5     A year is a leap year if it is divisible by 4,
6     except for end-of-century years, which must be divisible by 400.
7
8     Args:
9         year (int): The year to check.
10    Returns:
11        bool: True if the year is a leap year, False otherwise.
12    """
13    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
14        return True
15    else:
16        return False
17
18 # Example usage:
19 year = 2024
20 if is_leap_year(year):
21     print(f"{year} is a leap year.")
22 else:
23     print(f"{year} is not a leap year.")
```

2024 is a leap year.

[Note] exited with code=0 in 0.449 seconds

[Testing] python -u "C:\Users\pooja\AppData\Local\Temp\TempCodeRunner\11a.py" 2000 is a leap year.

[Note] exited with code=0 in 0.167 seconds

[Testing] python -u "C:\Users\pooja\AppData\Local\Temp\TempCodeRunner\11a.py" 1900 is not a leap year.

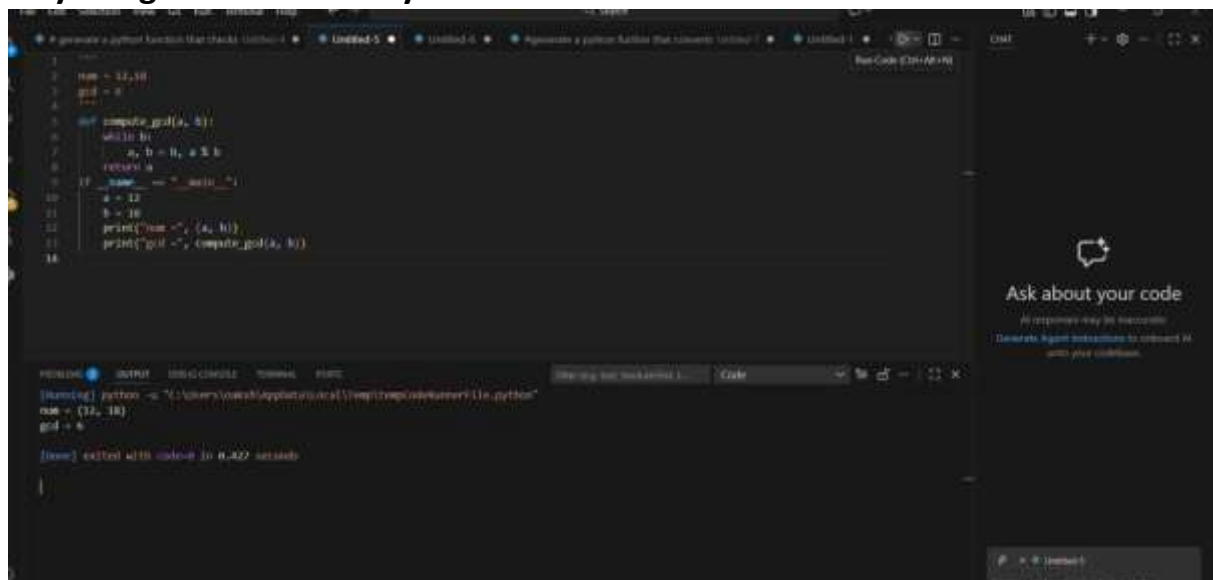
[Note] exited with code=0 in 0.161 seconds

Question 2: One-Shot Prompting (GCD of Two Numbers) Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6 Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency.



The screenshot shows a code editor with a Python function `compute_gcd(a, b)` that implements the Euclidean algorithm. The function uses a `while` loop to repeatedly replace `a` with `b` and `b` with `a % b` until `b` is zero, then returns `a`. Below the code, the output of running the script is shown: `nam = (12, 18)` and `gcd = 6`. The execution time is noted as 0.422 seconds.

```
1 nam = (12, 18)
2 gcd = 0
3
4
5 def compute_gcd(a, b):
6     while b:
7         a, b = b, a % b
8     return a
9
10 if __name__ == "__main__":
11     a = 12
12     b = 18
13     print('nam =', (a, b))
14     print('gcd =', compute_gcd(a, b))
15
```

Running: python -u "C:\Users\yoush\AppData\Local\Temp\codeinterpreter\11e.py"
nam = (12, 18)
gcd = 6

[jupyter] exited with output in 0.422 seconds

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21 Task:
- Examine how examples guide formula selection.
- Test edge cases.

```

1 # generate a python function that checks if a number is even or odd
2
3 def lim(a, b):
4     if a > b:
5         greater = a
6     else:
7         greater = b
8
9     while True:
10         if greater < a == 0 and greater < b == 0:
11             lim = greater
12             break
13         greater += 1
14
15 return lim

```

```

[Running] python -u "C:\Users\user\AppData\Local\Temp\code-runner\file.py"
12
[Done] exited with code:0 in 0.433 seconds

```

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion) Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

- Test with valid and invalid binary inputs.
- Identify missing validation logic.

```

1 # generate a python function that converts a binary number to decimal
2
3 def binary_to_decimal(binary_str):
4     decimal_value = 0
5     binary_str = binary_str[::-1] # Reverse the string to process from least significant bit
6     for index, digit in enumerate(binary_str):
7         if digit == '1':
8             decimal_value += 2 ** index
9     return decimal_value
10
11 # Example usage:
12 binary_number = "1111"
13 decimal_number = binary_to_decimal(binary_number)
14 print(f"The decimal value of binary {binary_number} is {decimal_number}")
15

```

```

[Running] python -u "C:\Users\user\AppData\Local\Temp\code-runner\file.py"
The decimal value of binary 1111 is 15
[Done] exited with code:0 in 0.383 seconds

```

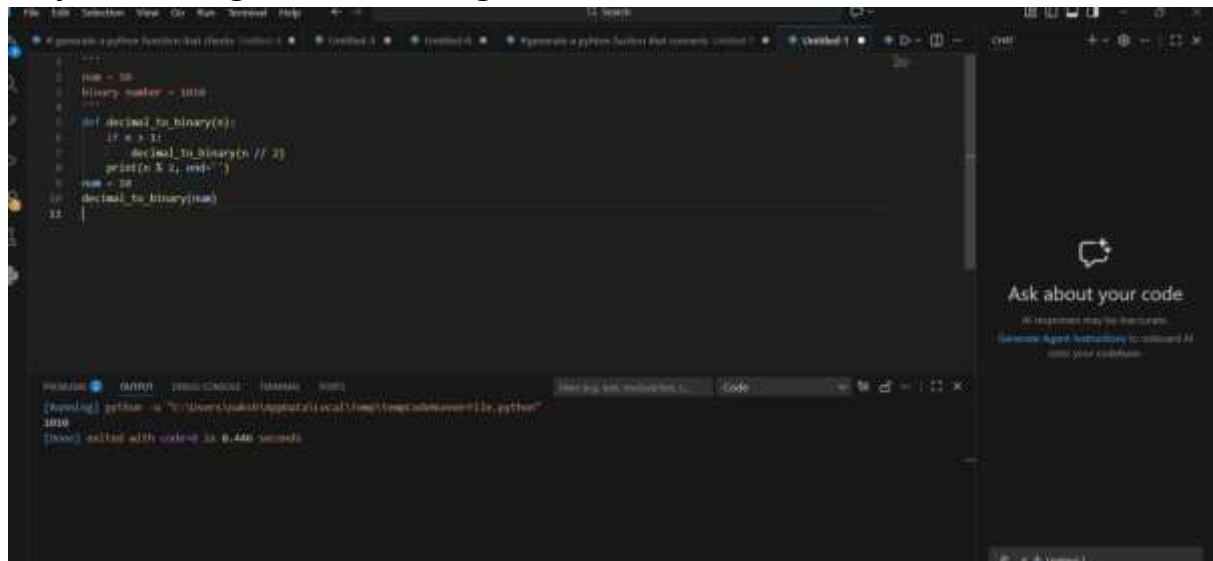
Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010 Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.



The screenshot shows a code editor with a Python script. The script defines a function `decimal_to_binary(n)` that recursively converts a decimal number to binary. The function uses integer division by 2 and modulo 2 to build the binary string. The script is executed, and the output is displayed in the console.

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